1.) Let g be a differentiable function.

(a) Use the definition of the derivative to compute the derivative of the function $\frac{1}{g}$, where it is defined. Check your answer with your instructor before moving on.

(b) Use the Product Rule together with your answer to part (a) to compute the derivative of the function $\frac{f}{g}$. Simplify your answer and check with your instructor before moving on. This is called the *Quotient Rule*.

2.) The Quotient Rule can be used to extend the Power Rule to the case where the exponent is a negative integer. Prove that if n is a positive integer, then $\frac{d}{dx}(x^{-n}) = -nx^{-n-1}$.

3.) Calculate the derivative of each of the following functions, and state where each is differentiable.

(a) $f(x) = \frac{1}{x}$.

(b) $g(t) = \frac{6}{t^3}$.

4.) Differentiate $y = \dfrac{x^2+x-2}{x^3+6}$.

5.) Differentiate $F(x) = \dfrac{3x^2 + 2\sqrt{x}}{x}$ in two ways. Which one is easier?

6.) Find the equation of the tangent line to the curve $y = \dfrac{\sqrt{x}}{1+x^2}$ at the point $(1, \frac{1}{2})$.

7.) Calculate the derivative of each of the remaining trig functions using the Quotient Rule. Simply each as much as possible.

(a) $\tan(x)$

(b) $\cot(x)$

(c) $\sec(x)$

(d) $\csc(x)$